CSC490 A3

Group 1 AGD-G
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Part 1

A core design principle of Nanochat is to remove unnecessary complexity and to select methods proven to improve the performance of small models. Our work extends this philosophy by evaluating methods overlooked in the recent February version of Nanochat. These three proposed architecture changes, derived from recent literature, are described below.

New Method	Technical Details	Motivation	Hypothesis Effect
No Positional Embedding (NoPE)	Traditionally, it is thought that positional information must be explicitly added to embeddings. However, Kazemnejad et al. (2023) demonstrated that decoder-only Transformers can effectively operate without explicit positional information because the causal attention mask inherently captures order.	Nanochat currently uses Rotary Position Embeddings (RoPE). Removing positional embeddings entirely could further simplify the architecture and improve model speed.	It is likely to have to no negative effect evaluations the context lengths the model was trained on, while potentially offering improved performance on length generalization beyond those training lengths. Additionally, this method trains more efficiently due to the removal of positional computation overhead.

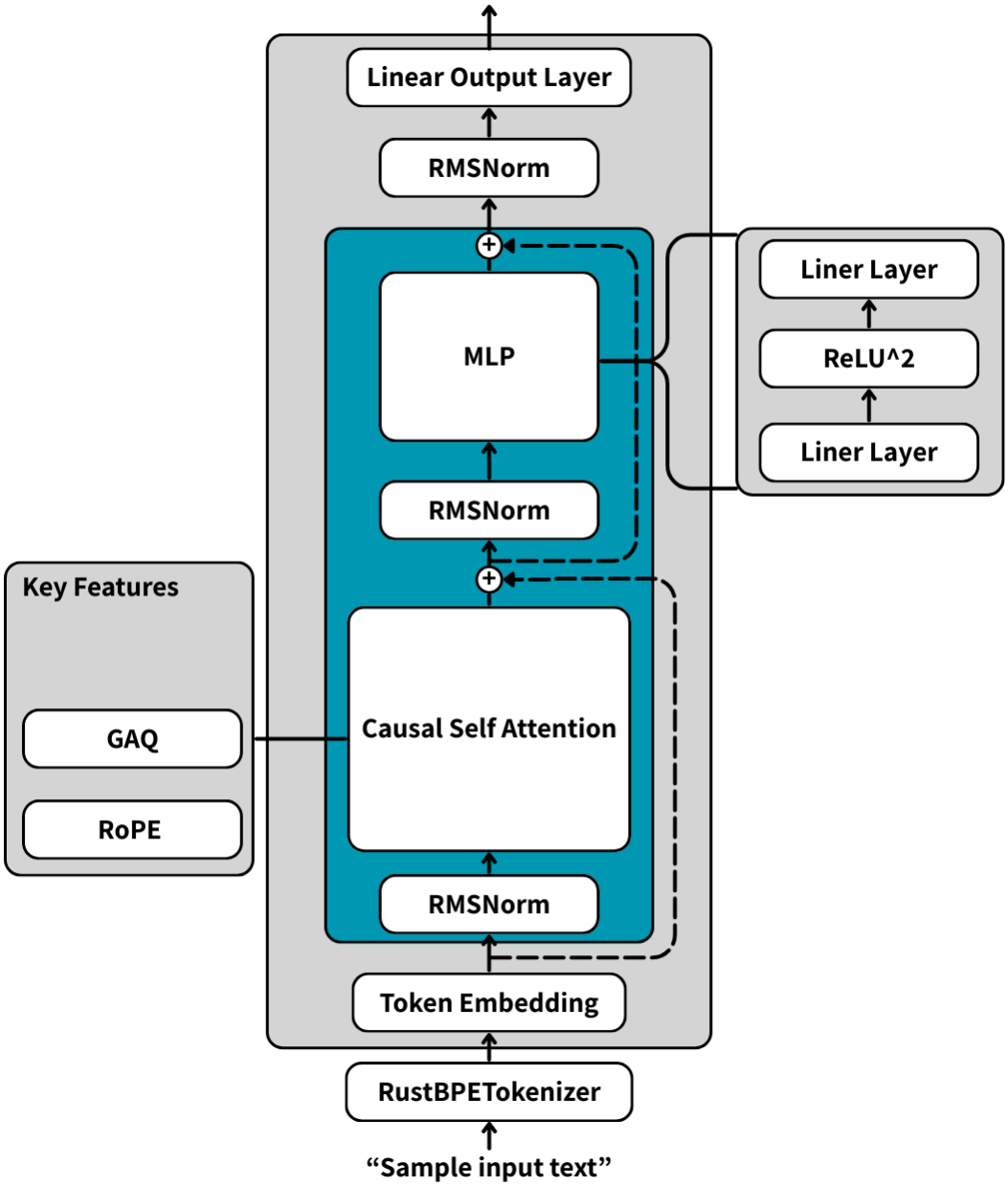


Diagram of Oct. 13th Nanochat Architecture

Affine Attention	Bae et al. (2026) in discovered that the standard softmax attention mechanism forces a strict unit-sum constraint. Affine attention modifies this by multiplying the softmax output by an input-dependent scaling factor (using a linear clipping activation function) and adding a bias term. The bias term is equal to the difference between the running average of the scaling factor divided by the key sequence length.	Bae et al. (2026) in recently discovered that the strict requirement for softmax attention to sum to 1 limits a model's flexibility. They found that adding gates and dynamically assigning head-specific scaling factors improves model performance. This shows how Nanochat's attention mechanism is no longer cutting edge.	This method will likely improve performance across a diverse array of tasks by enhancing attention flexibility and reducing first-token bias. There is uncertainty whether this method is as beneficial for non-distilled or smaller scale models like our Picochat. It will also incur a slight efficiency penalty due to the additional affine computations.
Multi-Head Latent Attention (MLA)	To improve efficiency, MLA compresses the multi-head attention mechanism. It uses a low-rank approximation to project the Keys and Values.	Nanochat currently uses Group-Query Attention, which sacrifices some accuracy for performance. With MLA, this tradeoff is mitigated, as shown in the DeepSeek V2 paper.	The model is predicted to show improvements in both performance and accuracy due to the compressed multi-headed attention.

Part 2

We decided to implement NoPE and Affine-Scaled Attention. These methods offered a unique opportunity to test cutting edge practices, with NoPE showing strong potential for length generalization and Affine Attention showing promise for training stability.

Training Setup

Our baseline configuration for Picochat was a model of depth 12. This size was chosen as it was the largest model we could train at a reasonable cost. Choosing a larger base model requires less extrapolation when predicting the performance of the full size depth 24 Nanochat. The training utilized 240 shards of data and a device batch size of 16, which is the recommended setting for the H100 to minimize the risk of out of memory errors. We utilized 8xH100 GPUs to ensure comparability to the base full size Nanochat. Flash Attention was employed to speed up processing without affecting accuracy. Mixed precision (FP8) was deliberately excluded to limit potential sources of error during development.

	CORE Score	Validation BPB (lower is better)	Cost (USD)	Training Time (sec)	MFU	Token/Sec
Base Picochat	0.1317	0.9032	$2.62	298.8	37.87	3,735,000
Affine Picochat	0.1390	0.9035	$2.98	339.8	31.82	3,137,000
NoPE Picochat	0.0978	0.9352	$2.56	291.7	37.47	3,696,000

Commentary

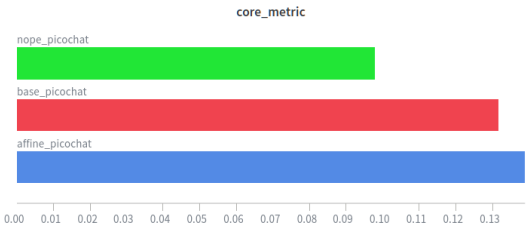
When comparing the Base and Affine Picochat models, we observed a 5.54% increase in the CORE Score, which jumped from 0.1317 to 0.1390. The CORE Score averages 22 tasks across a wide range of areas and serves as the primary evaluation metric for Nanochat. While Affine performed slightly worse on the validation bits-per-byte (BPB) metric with a nominal 0.03% increase, BPB is a removed measure of real world utility. Thus, the CORE Score is a more reliable indicator of downstream impact. These are highly promising results for Affine's ability to improve model performance, corroborating the findings of Bae et al. (2026) that Affine Attention enhances downstream task accuracy.

As expected, the Affine model demands more resources. It takes 13.72% longer to train (an additional 41 seconds) and processes text 16.01% slower (dropping from 3,735,000 to 3,137,000 tokens per second). Because cost is directly proportionate to training time, the Affine model is 13.74% more expensive. However, we anticipate these performance gains to scale well to larger models. A supplementary test controlling for identical training times still showed higher CORE scores for Affine, confirming that the additional compute cost is efficiently spent.

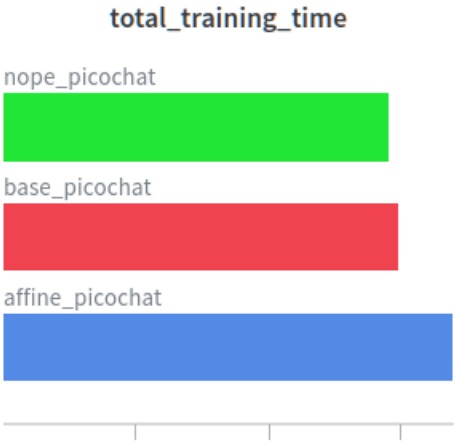
Conversely, the NoPE model yielded a 2.38% decrease in training time and a 2.29% lower training cost. However, it suffered a substantial 25.74% drop in the CORE score (falling from 0.1317 to 0.0978) alongside a 3.54% increase in validation loss. This appears to contradict the performance improvements noted by Kazemnejad et al. (2023) in The Impact of Positional Encoding on Length Generalization in Transformers. Deeper insight reveals that Kazemnejad et al. (2023) specifically highlighted NoPE's superiority in length generalization tasks (extrapolating to sequence lengths longer than seen in training) rather than standard in-distribution performance. The degraded CORE performance in our Picochat might simply reflect the model's small capacity or the lack of specific length extrapolation evaluations in the standard CORE suite. It is also possible that larger model sizes are required for NoPE to implicitly learn relative positional attention patterns effectively.

Results

By comparing baseline picochat model against affined-attention and NoPE ablations, we found that picochat with affined-attention performed the best overall.



Core metrics between models with ablations vs. Baseline



Training time between models with ablations vs. Baseline

Part 3

Method

We trained a two-staged version of Nanochat to test context-length continuation

Stage 1 (Checkpoint 1): short-context Pretraining.
Max sequence length: 512
Training endpoint checkpoint: step 500 (model_000500.pt)

Stage 2 (Checkpoint 2): long-context continuation (resumed from checkpoint1)
Increased max sequence length to 2048
Continued training to step 1000 (model_001000.pts

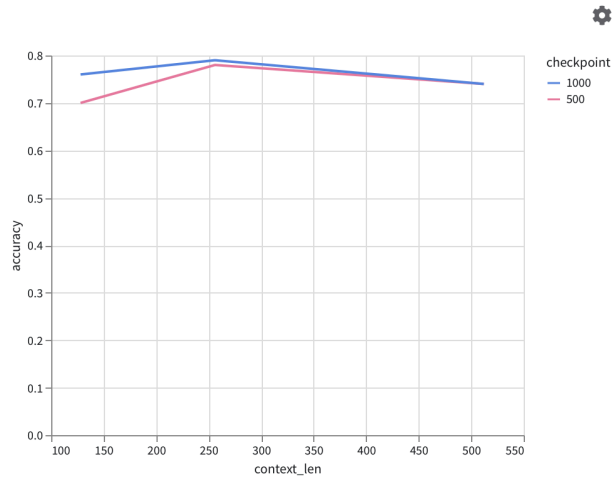
Justification

- Training from shorter to longer contexts is commonly used to improve stability and reduce early training cost while still enabling long-context capability later.
- Scaling-law and training-efficiency research supports training at shorter context to improve token throughput before paying full long-context cost

We built custom checkpoint-comparison evals (non-default Nanochat evals), focusing on checkpoint 1 vs checkpoint 2

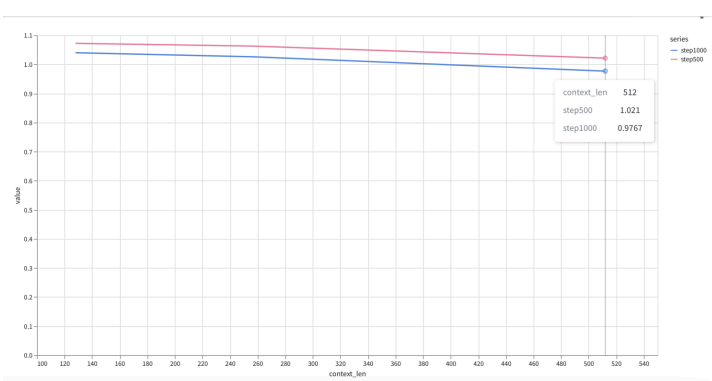
Context Lengths tested: $128, 256, 512$.

Custom Eval 1: Next-token pattern task. This task checks whether the model continue a simple repeating token pattern. Eg. a b c a b (correct output would be c).



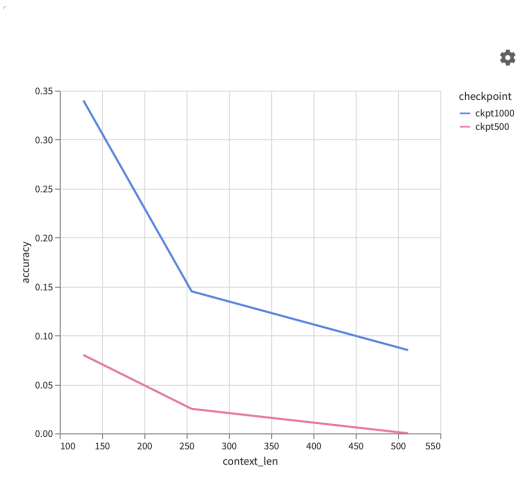
context_len	checkpoint_500 (BPB)	checkpoint_1000 (BPB)	delta (1000-500)
128	1.072253	1.039947	-0.032306
256	1.062778	1.026075	-0.036704
512	1.021409	0.976677	-0.044732

Custom Eval 2: Bits Per B-by-context curve. This eval measures validation bits-per-byte (BPB) at different context lengths (128, 256, 512). BPB is a compression-style language modeling metric, where lower is better. It checks whether the continued 2048-context checkpoint models text more efficiently than the 512-context checkpoint at shared lengths.



context_len	checkpoint_500 (acc)	checkpoint_1000 (acc)	delta (1000-500)
128	0.080	0.340	+0.260
256	0.025	0.145	+0.120
512	0.000	0.085	+0.085

Custom Eval 3: Induction probe. This eval checks context retrieval: place an earlier token pair (x, y) in the prompt, end with x, and test whether the model predicts y next. Accuracy measures how often retrieval succeeds. It probes whether longer-context continuation improved pattern lookup from earlier context.



task	context_len	checkpoint_500	checkpoint_1000	delta (1000-500)
abc	128	0.700	0.760	+0.060
abc	256	0.780	0.790	+0.010
abc	512	0.740	0.740	+0.000

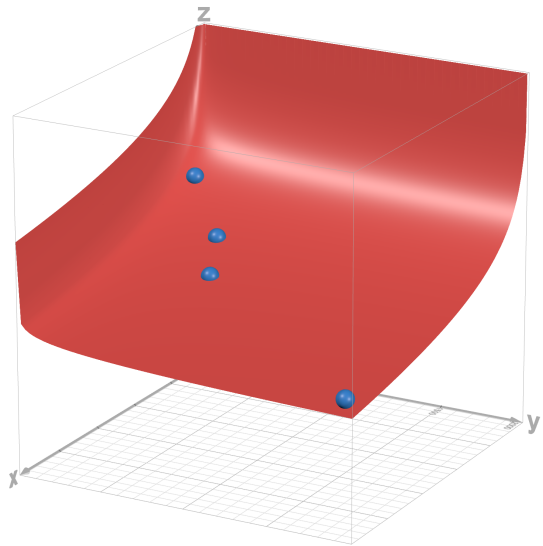
Part 4

Our Nanochat had a depth of 24 the same as the GPT-2-equivalent Nanochat. All other perimeters are the same as Picochat to allow for comparison.

Scaling Laws

Introducing the Chinchilla-Style Scaling Law: To analyze how model performance scales with model size and dataset size, we utilized a Chinchilla-style scaling law. This formulation states that validation loss (measured in bits-per-byte, or bpb) decreases as either model capacity or training data increases, following an inverse-power law where marginal improvements diminish at scale.
Approximation and Regression from Three Trial Runs: Given our constraint on the number of training runs, we applied a first-order approximation using log variables to create a locally linear model. Because depth is the main architectural change across our models, we substituted model depth (d) for parameter count (N). We trained three smaller Affine nanochat models at depths of 4, 8, and 12 to gather empirical data. Using these three trial runs, we fitted the regression equation: $L(d, D) \approx 2.2580 - 0.2280 \ln d - 0.03905 \ln D$. Both coefficients are negative, confirming the expected behavior that increasing either variable reduces validation loss.

Visualizing the Scaling Law: This scaling relationship is visually represented in the accompanying 3D surface graph. The graph illustrates the fitted regression model by plotting validation loss along the Z-axis against model depth and dataset size on the X and Y axes. The steep initial curve that gradually flattens out visually demonstrates how performance gains slow down as the model scales up.



Figure

Predicting the Full Nanochat Model: We used this fitted empirical regression to predict the performance of our full-size nanochat model. The final model has a depth of 24 and processes $1{,}916{,}796{,}928$ total training tokens over $7{,}312$ steps. Substituting these values into our equation ($d = 24$ and $D = 1{,}916{,}796{,}928$) yielded a predicted validation bpb of approximately 0.699.

Comparing the Existing Result: When we trained the actual depth-24 affine nanochat, it achieved a validation bpb of 0.7537. The regression model was slightly optimistic, overestimating the improvement by 0.055 bpb, which represents roughly a $7.3\,\%$ error relative to the true outcome. While the prediction captured the correct qualitative trend of improved performance at scale, the quantitative gap likely stems from limited data points. We only had three small runs to fit the model, depth is merely a proxy for exact parameter count, and larger models often exhibit different optimization dynamics than their smaller counterparts. Despite these limitations, the scaling-law analysis proved highly useful by confirming that increasing model capacity and dataset exposure substantially improves language modeling quality over the picochat models.

To analyze how model performance scales with both model size and dataset size, we trained several NanoChat models with affine attention at different depths and measured their validation

performance. The goal is to use the smaller pico-scale runs to estimate a scaling relationship and use it to predict the performance of the full NanoChat model with depth 24.

Emergent Abilities

The Nanochat model (depth=24) showed emergent abilities over the Picochat model (depth=12). The Nanochat model was able to answer all of the following questions (per our judgement) while the Picochat model was not able to answer any of them.

- Name the primary colours.
- How many continents are there?
- What is "Hello" in French?
- Who was the first prime minister of Canada?
- What is the heaviest metallic element?
- Who should I visit to get a cavity filled?
- How many wheels does a bicycle have?
- Is the sun or the moon bigger?
- What year was American independence?
- What is the capital city of Canada?

Full responses for both models can be found in the appendix.

We found that factual questions like these were the best at differentiating the models. Both models were able to answer coding questions, with answer quality depending on the question, not the model. Neither model was capable of mathematical nor logical reasoning, with both models failing on "What is 1 + 1?" and "Which is heavier, a pound of feathers or a pound or stone?" This is likely because we did not run the reinforcement learning step, which is focused on math. Both models were able to count occurrences of letters in words, since the "spelling bee" evaluations were used in training, but both failed to produce rhyming words.

Takeaways and improvements

	CORE Score	Validation BPB (lower is better)	Cost (USD)	Training Time (Hour)
Base Nanochat	0.2585	0.7483	~$96.06*	3.04
Affine Nanochat	0.2548	0.7537	$89.43	2.83

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* Using Modal Pricing for H100 GPUs

Ultimately, we did not achieve the improvements to accuracy that we hypothesized from our literature review. By selecting No Positional Embedding (NoPE) and Affine-Scaled Attention, we took a calculated risk on less proven, experimental methods for architectural improvement. Our research demonstrates that the efficacy of both methods is heavily dependent on the surrounding architectural choices, evaluation metrics, and model scale.

Several factors likely contributed to our pico-scale results diverging from the established literature. The most likely factor is the difference in evaluation metrics. The choice of benchmark directly influences the perceived success of any modification. Furthermore, when we scaled our architectural modifications up to a larger model parameter count to compare against the Standard Nanochat baseline, the larger experimental model underperformed the standard baseline. This occurred despite the promising 5.54% CORE score improvement observed in our pico-scale Affine model. This inversion highlights a critical challenge in architectural research: performance gains at the pico-scale do not strictly guarantee monotonic improvements at scale. It is important to note the base nanochat model was slightly overtrained to make it more comparable to GPT-2. This could be a factor in the difference of results.

The Standard Nanochat baseline is the product of extensive hyperparameter tuning (e.g., learning rate schedules, weight decay, initialization variances) specifically optimized for standard softmax attention. Introducing dynamic scaling factors (Affine) or removing explicit positional cues (NoPE) fundamentally alters the loss landscape and gradient flow. Utilizing the baseline's hyperparameters for our modified architecture likely forced the model into a suboptimal optimization trajectory. At parameters, the model is significantly more sensitive to these hyperparameter mismatches than the 12-layer picochat.

To better understand these discrepancies, future work should first apply the identical evaluation suites used in the original papers to rule out benchmark bias. Subsequent tests could then align our baseline architecture closer to the models featured in the literature, such as utilizing a distilled model for NoPE or testing non-multihead attention for Affine. If these adjustments yield the expected performance gains, it would confirm that NoPE and Affine are tightly connected to these specific configurations.

These scaling results reinforce the complexity of language model development. While exploring novel mechanisms like Affine Attention or NoPE is theoretically sound, their successful implementation at the billion-parameter scale requires bespoke hyperparameter sweeps and extended training horizons. Despite the mixed outcomes of these specific ablations, Multi-Head Latent Attention remains a highly promising method for future improvements to nanochat.

Citations

Chen, S., Wong, S., Chen, L., & Tian, Y. (2023, June 28). Extending context window of large language models via positional interpolation. arXiv.org. https://arxiv.org/abs/2306.15595

DeepSeek-AI, Liu, A., Feng, B., Wang, B., Wang, B., Liu, B., Zhao, C., Dengr, C., Ruan, C., Dai, D., Guo, D., Yang, D., Chen, D., Ji, D., Li, E., Lin, F., Luo, F., Hao, G., Chen, G., … Xie, Z. (2024, June 19). *Deepseek-V2: A strong, economical, and efficient mixture-of-experts language model*. arXiv.org.

Haviv, A., Ram, O., Press, O., Izsak, P., & Levy, O. (2022, December 5). Transformer language models without positional encodings still learn positional information. arXiv.org. https://arxiv.org/abs/2203.16634

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., Casas, D. de L., Hendricks, L. A., Welbl, J., Clark, A., Hennigan, T., Noland, E., Millican, K., Driessche, G. van den, Damoc, B., Guy, A., Osindero, S., Simonyan, K., Elsen, E., … Sifre, L. (2022, March 29). *Training compute-optimal large language models*. arXiv.org. https://arxiv.org/abs/2203.15556

Jeongin Bae Baeseong Park Gunho Park Minsub Kim Joonhyung Lee Junhee Yoo Sunghyeon Woo Jiwon Ryu Se Jung Kwon Dongsoo Lee, Bae, J., , Park, B., Park, G., Kim, M., Lee, J., Yoo, J., Woo, S., Ryu, J., Kwon, S. J., Lee, D., Agarwal, S., Bahdanau, D., Barbero, F., Bisk, Y., Choromanski, K., Clark, C., Clark, P., … Zellers, R. (n.d.). *Affine-Scaled Attention: Towards Flexible and Stable Transformer Attention*. Affine-scaled attention: Towards flexible and stable transformer attention. https://arxiv.org/html/2602.23057v1

Appendix

Question-Answering Test

Answers from Picochat with Affine-Scaled-Attention

==========> Name the primary colours. <==========
The primary colours of a color are:
Red, Blue, Green, Yellow, Black, Blue, Red, Red, Green, Blue, Red, Red, Green, Blue, Red, Green, Blue, Red, Red, Green, Blue, Red, Red, Green, Blue, Red, Red, Green, Blue, Red, Green, Blue, Red, Red, Green, Blue, Red, Green, Blue, Red, Green, Blue, Red, Green, Red, Green, Red, Green, Blue, Red, Red, Green, Red, Green, Blue, Red, Green, Blue, Red, Red, Green, Blue, Red, Green, Blue, Red, Green, Red, Red, Green, Blue, Red, Red, Green, Blue, Red, Red, Green, Blue, Red, Red, Green, Red, Green, Blue, Red, Blue, Red, Green, Blue, Red, Red, Green, Red, Green, Blue, Red, Green, Red, Green, Blue, Red, Blue, Red, Green, Blue, Red, Green, Blue, Red, Green, Blue, Red, Green, Blue, Red, Green, Blue, Red, Green, Blue, Red, Green,
==========> How many continents are there? <==========

There are 27 known continents, which are the 20 major supercontinents. The remaining 19 are the 19 smaller ones, which include the 5 continents that are part of the Earth's larger sphere.<|assistant_end|>
==========> What is "Hello" in French? <==========
In French, "Hello" is a greeting used to greet someone in a relationship. It is a greeting that is commonly used in everyday conversations and situations where a person is in a relationship. The word "Hello" is often used in phrases such as "Good day," "Good evening," or "Good evening." It is also used in place of "friend" to indicate that the person is in a relationship.<|assistant_end|>
==========> Who was the first prime minister of Canada? <==========
The first prime minister of Canada was Justin Trudeau, who was born on January 1, 1901. Trudeau was a member of the Canadian Parliament and was known for his ability to negotiate with the Canadian government and promote the Canadian government. He was a member of the Canadian National Parliament and was also a member of the Canadian Parliament. Trudeau's leadership was instrumental in shaping Canadian politics and culture.
Trisha Léves, the first prime minister of Canada, was also born on January 1, 1901, and was known for her strong stance on the Canadian government and her commitment to promoting the Canadian government. Léves was a member of the Canadian Parliament and was also a member of the Canadian Parliament. Léves's leadership was instrumental in shaping the Canadian government's policies and actions.
As the first prime minister of Canada, Trudeau played a crucial role in shaping Canadian politics and culture, and his legacy continues to influence Canadian politics and society today.<|assistant_end|>
==========> What ingredients do I need for pancakes? <==========
Pancakes are typically made with a variety of ingredients that can be used to create delicious and complex pancakes. One of the main ingredients is a mix of flour, sugar, eggs, milk, and water. Flour is the backbone of pancakes, providing the structure and texture of the dough. Sugar, which is added to the batter, helps to create a soft and chewy texture.
Other key ingredients include butter, which is a natural sweetener, and eggs, which are a quick source of protein. Milk, which provides a good source of calcium and vitamins, is also essential for pancakes. Water, which is essential for maintaining the texture and flavor of the pancakes.
To make pancakes, you can use a variety of flour and sugar, such as wheat flour, rice flour, and corn flour. You can also use eggs, which are a natural source of protein and fat. Milk, which comes from milk but is not as natural as eggs, is another option. Water, which is used to make pancakes, is also a crucial ingredient. The right amount of water can make all the difference in the texture and flavor of pancakes.<|assistant_end|>
==========> What is the heaviest metallic element? <==========
The heaviest metallic element is copper, with a weight of 600 g. It is often used as a reference point for various applications, including alloys, polymers, and other materials. Copper's high density and strength make it a key component in many alloys, such as stainless steel, aluminum, and ceramics.<|assistant_end|>
==========> Who should I visit to get a cavity filled? <==========
When it comes to cavity filled, the decision to visit depends on your personal preferences, lifestyle, and what you hope to achieve. Here are some factors to consider:
If you're looking for a more substantial boost, you might want to consider visiting a cavity filled in a few months' worth of living costs. This could be a significant investment, especially if you have family or other commitments.
However, if you're on a tight budget or have access to affordable housing, a cavity filled in a few months' worth of living costs could be a worthwhile investment. This might mean you can afford a larger home or a smaller space, or even a smaller apartment.
Another factor to consider is your financial situation. If you're not financially prepared for the investment, a cavity filled in a few months' worth of living costs might not be worth it. On the other hand, if you're financially stable and able to afford the initial investment, a cavity filled in a few months' worth of living costs could be a worthwhile investment.

Ultimately, the decision to visit a cavity filled in a few months' worth of living costs depends on your individual circumstances and priorities. It's essential to weigh the pros and cons carefully and consider what will work best for you and your family
==========> How many wheels does a bicycle have? <==========
A bicycle has 8 wheels, which are essentially the wheels of a bicycle tire. This number is based on the principle of tire rotation, which involves rotating a wheel by 360 degrees (or 60 degrees) around a central axis.<|assistant_end|>
==========> Is the sun or the moon bigger? <==========
The size of the Earth and the moon can be subject to various factors, particularly in the context of space travel and space exploration. However, based on astronomical observations and historical records, it's difficult to pinpoint the exact size of the Earth and the moon.
The moon is about 40% larger than the Earth, with a diameter of approximately 300 kilometers. This is because the moon is positioned about 2.6 times larger than the Earth in terms of distance from the Sun, making it larger in diameter.
Additionally, the Earth and the moon are not always in a perfect elliptical orbit, which can affect their orbits. For example, the Earth is slightly tilted at an angle of about 23.5 degrees, while the moon is about 2.6 times larger. This tilts the moon's axis relative to the Earth's, a phenomenon known as tidal locking.
It's also worth noting that these factors can affect the size of the Earth and the moon. For instance, the Earth's diameter is approximately twice as large as the Moon's, and the moon's diameter is about 3 times larger.<|assistant_end|>
==========> What year was American independence? <==========
American independence was a period of American independence from the British colonies in the late 18th and early 19th centuries. It began in 1813 and lasted until 1852. The year was marked by a series of events and conflicts between the colonies and the British government, which ultimately led to the outbreak of the American Civil War in 1861. The war was sparked by the Battle of Waterloo, where Napoleon Bonaparte, a German general, was killed in a battle against the French army. The American Civil War was fought over a region of land called Napoleon's territory, and it lasted for 24 months, from March 1861 to April 1865. The war lasted for 6 months and ultimately resulted in the capture of 13 American Confederate states, including the Union of South Carolina and Virginia, and the eventual surrender of South Carolina to the Union.<|assistant_end|>
==========> What is the capital city of Canada? <==========
The capital city of Canada is Toronto, Canada. It is a vibrant and cosmopolitan city, situated in the northern part of the country and nestled in the heart of the Canadian wilderness. The city's unique blend of history, culture, and natural beauty makes it a popular destination for tourists and locals alike.<|assistant_end|>

Answers from Nanochat with Affine-Scaled-Attention

==========> Name the primary colours. <==========
The primary colours are a set of colours that are the building blocks of all other colours. The primary colours are red, blue, and yellow. These colours are derived from the additive colours, which are the primary colours themselves. They are named so because they can be created by adding other colours together in different proportions. The primary colours are often described as red, blue, and yellow because they are the fundamental colours used in many different types of art, design, and everyday life.<|assistant_end|>
==========> How many continents are there? <==========
There are 7 continents in total. These continents are: Africa, Antarctica, Asia, Australia, Europe, North and South America, and the Arctic and Antarctica.<|assistant_end|>
==========> What is "Hello" in French? <==========
In French, "Bonjour" is a greeting that means "Hello" or "Good day." It is commonly used in both formal and informal settings, and is often used in casual conversations, greetings, and

even in formal correspondence. It is a simple and straightforward way to start a conversation or introduce yourself.<|assistant_end|>
==========> Who was the first prime minister of Canada? <==========
The first prime minister of Canada was John A. Macdonald, who served from 1867 to 1869. He was appointed by Governor General Lord Stanley, who was the prime minister of Canada from 1865 to 1866. Macdonald was a prominent figure in Canadian politics during his time, serving as a member of the British North America Act and later as a member of the Conservative Party. He is often credited with shaping Canada's early years and signing the Canadian Charter Act in 1867.<|assistant_end|>
==========> What ingredients do I need for pancakes? <==========
For a classic pancake recipe, you'll need a few key ingredients. These include flour, eggs, butter or oil, salt, sugar, milk or milk alternatives, and a bit of vanilla extract if you're using butter. For a classic pancake recipe, you'll also need a mixing bowl, a spatula, and a mixing spoon.
Additionally, you'll need eggs and milk for the batter. The type of eggs you use is also important, as scrambled eggs are often used in pancakes, while fried eggs can add a bit of extra texture.
If you're using oil, you'll need to stir it into the batter before adding the eggs. If you're using milk, you'll need to add it to the eggs. If you're using milk alternatives like almond milk, coconut milk, or soy milk, you may need to adjust the ratio accordingly.
Once you have your ingredients, start by mixing the batter in a bowl until it's smooth and well combined. Then, use a spatula to gently fold in the eggs, mixing them in until they're evenly distributed. Finally, heat a skillet over medium heat and cook the pancake until golden brown and crispy on the outside, about 5-
==========> What is the heaviest metallic element? <==========
The heaviest metallic element is typically considered to be gold, with a density of approximately 19.3 grams per cubic centimeter (g/cm). However, gold is not the heaviest metallic element, as the heaviest element is often considered to be platinum, with a density of around 19.2 g/cm. Platinum is actually the heaviest naturally occurring element, making it a rare and precious metal.<|assistant_end|>
==========> Who should I visit to get a cavity filled? <==========
To get a cavity filled, it's generally recommended to visit a dentist or an endodontist, but more specifically, to visit an endodontist. This is because endodontists have specialized training in the diagnosis, treatment, and prevention of dental pulp diseases, including cavities. Endodontists are dentists who have completed additional training in endodontics, which includes the diagnosis, treatment, and prevention of endodontic diseases. They can perform root canal procedures and other endodontic treatments, such as apicoarachnidodentistry.
Visiting an endodontist is recommended for several reasons:
- They have more advanced training and equipment, which can help them diagnose and treat complex cases.
- Endodontists are experienced in treating endodontic diseases, which can be more challenging than general dentistry.
- They can perform procedures that are not available to general dentists, such as microsurgical endodontic procedures.
- They are often able to provide more comprehensive and personalized care, including treatment planning and follow-up.
If you're not already seeing an endodontist, consider scheduling an appointment with your dentist for a routine check-up. They can help ensure that you receive the necessary care and attention to prevent cavities.<|assistant_end|>
=========> How many wheels does a bicycle have? <==========
A bicycle typically has 2 wheels, but some high-performance or hybrid bicycles may have as few as 1 or as many as 3 wheels, depending on the design and specific requirements. The wheels are connected to the frame using a set of rods and bolts, which are designed to provide stability and secure the wheels to the ground.<|assistant_end|>

==========> Is the sun or the moon bigger? <==========
The size of the sun and the moon is a topic of debate among astronomers and scientists. The sun is about 91.4 million kilometers (56.4 million miles) in diameter, while the moon is about 238,000 kilometers (149,000 miles) in diameter. Although the moon is much smaller than the sun, its size is still significant, especially in comparison to the sun, which is about 100 times larger.
The reason for the moon's smaller size is due to its much smaller mass and density. The moon's gravity is only about one-sixth of the Earth's gravity, which makes it much weaker than the sun's gravity. This weaker gravity allows the moon to retain a significant amount of its original material, which is why it has no atmosphere and no liquid water on its surface.
In contrast, the sun's gravity is much stronger, making it much larger than the moon. While the sun is much larger than the moon, the moon's size is still significant in terms of its mass and volume. The moon's orbit around the Earth is also much closer to us than the sun's orbit around the Earth, which affects its size and the amount of heat it receives from the sun.<|assistant_end|>
==========> What year was American independence? <==========
American independence was declared by the Continental Congress on July 4, 1776. It was initially signed by 12 of the 13 colonies. The signing of the Declaration of Independence by the Continental Congress marked the official beginning of the United States of America.<|assistant_end|>
==========> What is the capital city of Canada? <==========
The capital city of Canada is Ottawa. Ottawa is the capital of the Canadian province of Ontario and the nation of Canada, and it serves as the federal capital. It is located in the northern part of the country, near the Great Lakes and the Ottawa River. Ottawa is also the capital of the Province of Ontario, which is part of the province of Ontario, Canada.<|assistant_end|>